



NATIONAL OPEN UNIVERSITY OF NIGERIA
University Village, Plot 91, Cadastral Zone, Nnamdi Azikiwe Expressway, Jabi, Abuja

FACULTY OF SCIENCES
DEPARTMENT OF MATHEMATICS
2022_2 Examinations

Course Code: MTH301
Course Title: Functional Analysis I
Credit Unit: 3
Time Allowed: 3 Hours
Total: 70 Marks
Instruction: Answer Question One (1) and Any Other 3 Questions

(1) (a) Define the followings:

- (i) convergent Sequence in a metric space **(3 Marks)**
- (ii) closed, open, infinite intervals on the real line. **(3 Marks)**

(b) Let (X, d) be a metric space. Prove that a subset A of X is closed in (X, d) if and only if every convergent sequence of points in A , converges to a point in A . In particular, prove that A is closed in (X, d) if and only if $a_n \rightarrow x$, where $x \in X$ and a_n is a sequence of points in A , for all n , implies that $x \in A$. **(9 Marks)**

(c) Let (X, d) and (Y, d_1) be metric spaces and f a mapping of X into Y . Let τ and τ_1 be the topologies determined by d and d_1 respectively. Prove that $f: (X, \tau) \rightarrow (Y, \tau_1)$ is continuous if and only if $x_n \rightarrow x \rightarrow f(x_n) \rightarrow f(x)$. That is if $x_1, x_2, \dots, x_n, \dots$ is a sequence of points in (X, d) converging to x , then the sequence of points $f(x_1), f(x_2), \dots, f(x_n), \dots$ in (Y, d_1) converges to $f(x)$. **(10 Marks)**

(2) (a) Define the followings:

- (i) a metrisable topological space **(3 Marks)**
 - (ii) Euclidean space **(3 Marks)**
 - (iii) Absolute value **(3 Marks)**
 - (iv) Norm of a vector. **(3 Marks)**
- (b) Let $x \in \mathbb{R}^n$ and $\varepsilon > 0$. Prove that the set $B(x, \varepsilon) = \{y \in \mathbb{R}^n: d(x, y) < \varepsilon\}$ is open. **(3 Marks)**

- (3) Define the following
- (i) an interior point (1 marks)
 - (ii) boundary point (2 marks)
 - (iii) closure of a subset of a set (3 marks)
 - (iv) accumulation point (3 marks)
 - (v) closed set (3 marks)
 - (vi) interior of a set. (3 Marks)
- (4) (a)(i) Totally bounded metric space (3 Marks)
- (ii) Sequentially compact metric space (3 Marks)
- (b) Let X be a metric space and let Y be a subspace of X . Prove that
- (i) If X is compact and Y is closed in X , then Y is compact. (4 Marks)
 - (ii) If Y is compact, then it is closed in X . (5 Marks)
- (5) a(i) Define a topological space (4 Marks)
- (ii) Give three examples of topological spaces (6 Marks)
- (b) Let (K, d_K) be a compact metric space. Let (Y, d_Y) be any metric space and let $f: K \rightarrow Y$ be continuous. Prove that $f(K)$ is compact. (5 Marks)
- (6) a(i) If (A, d_A) and (B, d_B) are metric spaces, when is a function $f: A \rightarrow B$ said to be continuous? (3 Marks)
- (ii) Let f and g be real-valued functions with $\text{domain}(f) = \text{Range}(g) = D \subset \mathbb{R}^N$. If $\lim_{x \rightarrow x_0} f(x) = l$ and $\lim_{x \rightarrow x_0} g(x) = m$ then state the three limit theorems concerning the sum, product and quotient of the above function. (6 Marks)
- (b) Given the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $(x_0, y_0) = (1, 3)$. Compute the limits of the following functions as $(x, y) \rightarrow (1, 3)$:
- (i) $\lim_{(x,y) \rightarrow (1,3)} f(x,y)$, if $f(x,y) = \frac{2x}{x^2 + y^2 + 1}$. (3 Marks)
 - (ii) $\lim_{(x,y) \rightarrow (1,3)} f(x,y)$, if $f(x,y) = x^2 + y^2 + 1$. (3 Marks)